

PES UNIVERSITY, Bangalore
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Department of Computer Science & Engineering
UE24CS341A : Software Engineering

LAB - 02

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Agile Backlog Creation & Sprint Simulation in Jira

Problem Statement #64: Disaster Relief Supply & Volunteer Coordinator

1. Backlog: Epics & User Stories

Epic 1: Shelter Supply Requisition Matching (→ FR-001)

Story 1.1: Auto-Match Requisitions

As a Disaster Manager

I want the system to automatically match incoming shelter supply requisitions against warehouse stock
So that I can quickly see what's available to dispatch.

Story 1.2: Flag Critical Priority

As a Disaster Manager

I want critical requisitions (water, medical) to be automatically flagged with the highest dispatch priority
So that life-critical needs are never delayed behind lower-priority items.

Story 1.3: Auto-Add to Manifest

As a Disaster Manager

I want matched requisitions added directly to the logistics manifest
So that I don't have to manually re-enter dispatch data.

Epic / Story	Priority	Points
1.1 Auto-Match Requisitions	High	5
1.2 Flag Critical Priority	High	3
1.3 Auto-Add to Manifest	Medium	2

Epic 2: Volunteer Skill Profile Management (→ FR-002)

Story 2.1: Create/Update Skill Profile

As a Relief Volunteer

I want to create and update my skill profile
So that managers know what tasks I am qualified for.

Story 2.2: Real-Time Skill Sync

As a Relief Volunteer

I want my updated skills to reflect in manager search results within seconds
So that I can be assigned to urgent tasks without delay.

Story 2.3: Search Volunteers by Skill

As a Disaster Manager

I want to search volunteers by skill type

So that I can quickly find the right person for a task.

Epic / Story	Priority	Points
2.1 Create/Update Skill Profile	Medium	3
2.2 Real-Time Skill Sync	Medium	5
2.3 Search Volunteers by Skill	Medium	2

Epic 3: Skill-Based Volunteer Dispatch (→ FR-003)

Story 3.1: Dispatch Volunteer to Task

As a Disaster Manager

I want to dispatch a volunteer to a shelter task

So that shelters receive the right help.

Story 3.2: Filter Candidates by Skill

As a Disaster Manager

I want the system to filter volunteer candidates by the skill a task requires

So that unqualified volunteers are never shown as candidates.

Story 3.3: Dispatch Notification

As a Relief Volunteer

I want to receive a notification when I am dispatched to a task

So that I know where to report.

Epic / Story	Priority	Points
3.1 Dispatch Volunteer to Task	High	5
3.2 Filter Candidates by Skill	High	3
3.3 Dispatch Notification	Low	1

Epic 4: Relief Vehicle Delivery Tracking (→ FR-004)

Description: Track the real-time location and status of relief vehicles.

Story 4.1: Real-Time Vehicle Map

As a Disaster Manager

I want to see the real-time location of relief vehicles on a map

So that I can monitor deliveries.

Story 4.2: Auto Status Updates

As a Disaster Manager

I want vehicle delivery status to update automatically (Pending / In-Transit / Delivered)

So that I always have accurate delivery information.

Story 4.3: Geofence Delivery Confirmation

As a Disaster Manager

I want a vehicle's status to change to "Delivered" automatically when it enters a shelter's geofence

So that I don't have to manually confirm each delivery.

Epic / Story	Priority	Points
4.1 Real-Time Vehicle Map	High	8
4.2 Auto Status Updates	High	5
4.3 Geofence Delivery Confirmation	Medium	3

Epic 5: Delivery Delay Alerts & Monitoring (→ FR-005)

Story 5.1: ETA Exceeded Alert

As a Disaster Manager

I want to be alerted when a vehicle exceeds its estimated delivery time

So that I can respond to potential problems quickly.

Story 5.2: Fast Alert Generation

As a Disaster Manager

I want alerts generated within 2 minutes of an ETA being exceeded

So that delays are caught almost immediately.

Story 5.3: Alert History Log

As a Disaster Manager

I want to view a log of past delivery alerts

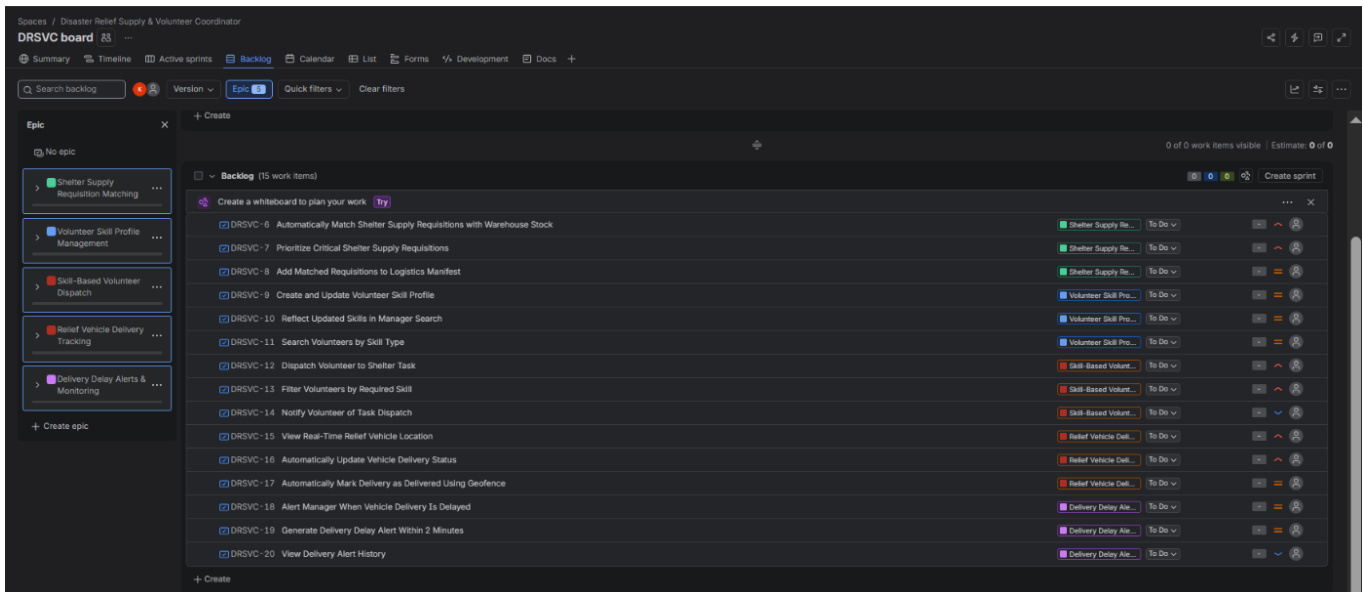
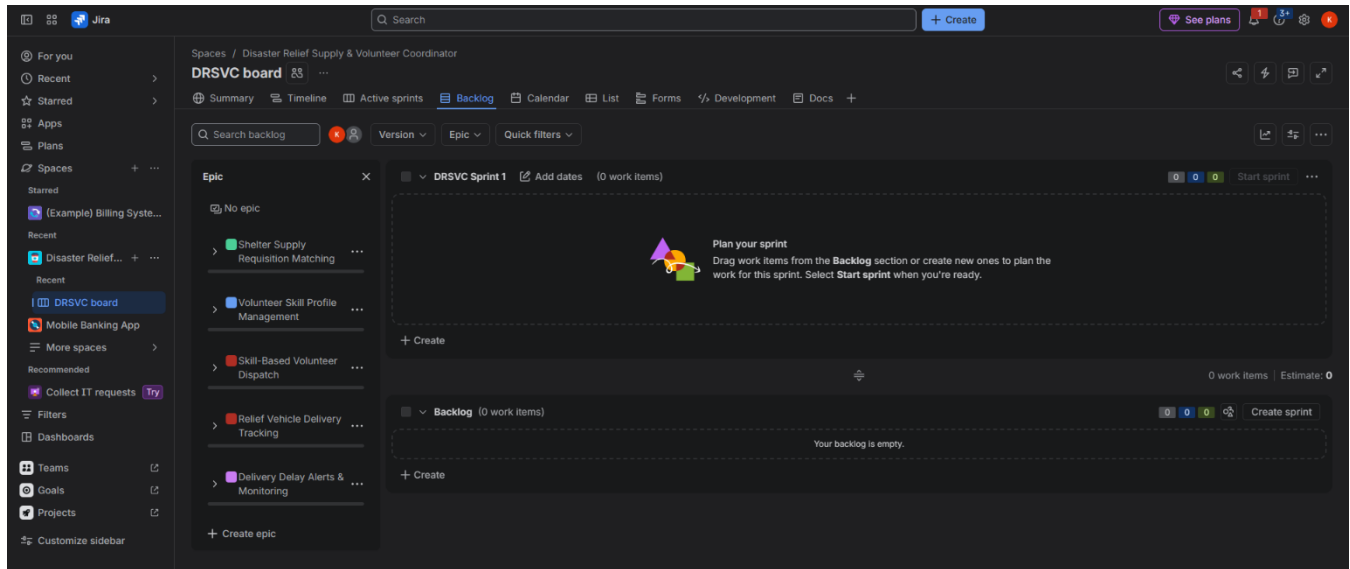
So that I can analyse recurring delivery issues.

Epic / Story	Priority	Points
5.1 ETA Exceeded Alert	Medium	3
5.2 Fast Alert Generation	Medium	2
5.3 Alert History Log	Low	1

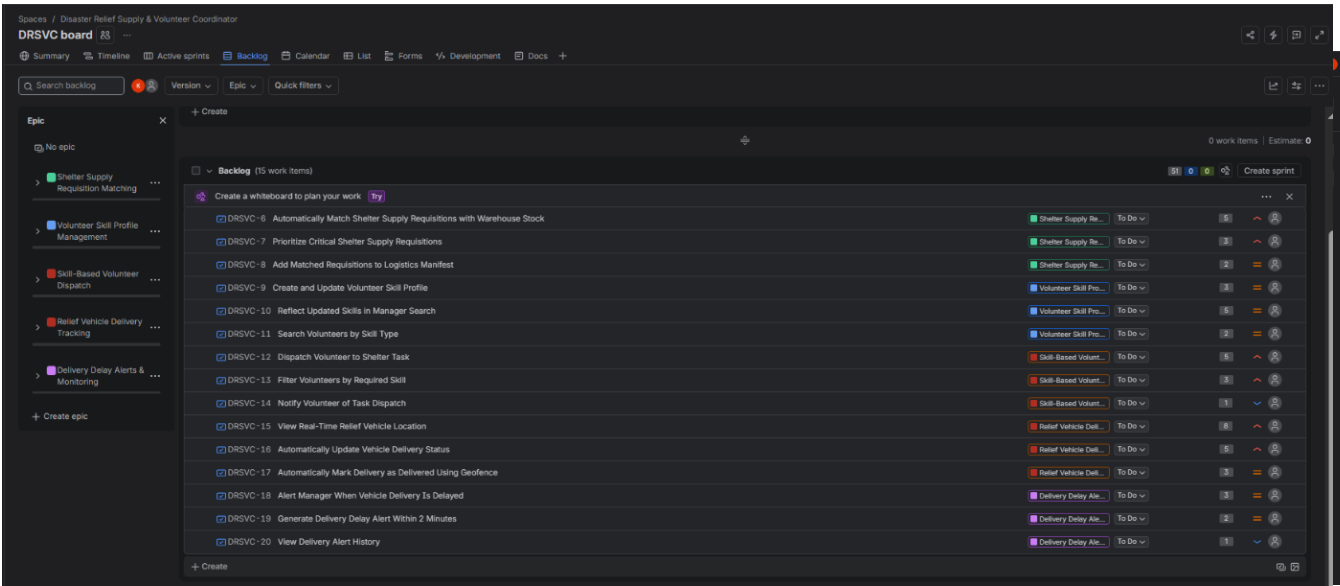
Total: 51 story points across 15 user stories.

2. Required Screenshots

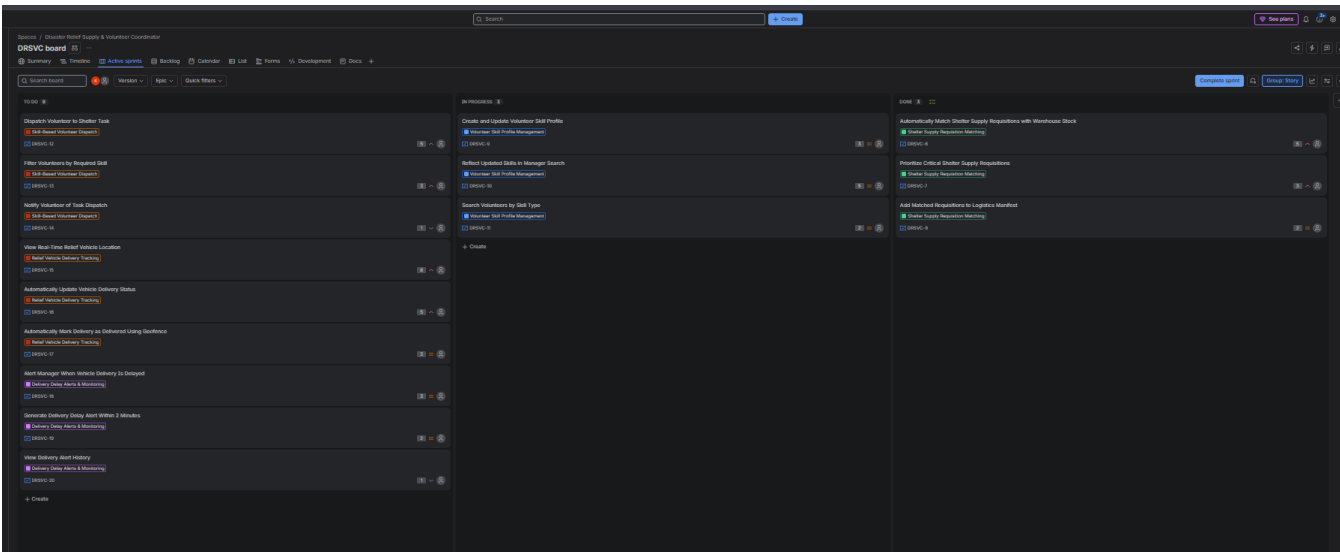
2.1 — Jira Backlog (all 5 Epics with their User Stories nested underneath)

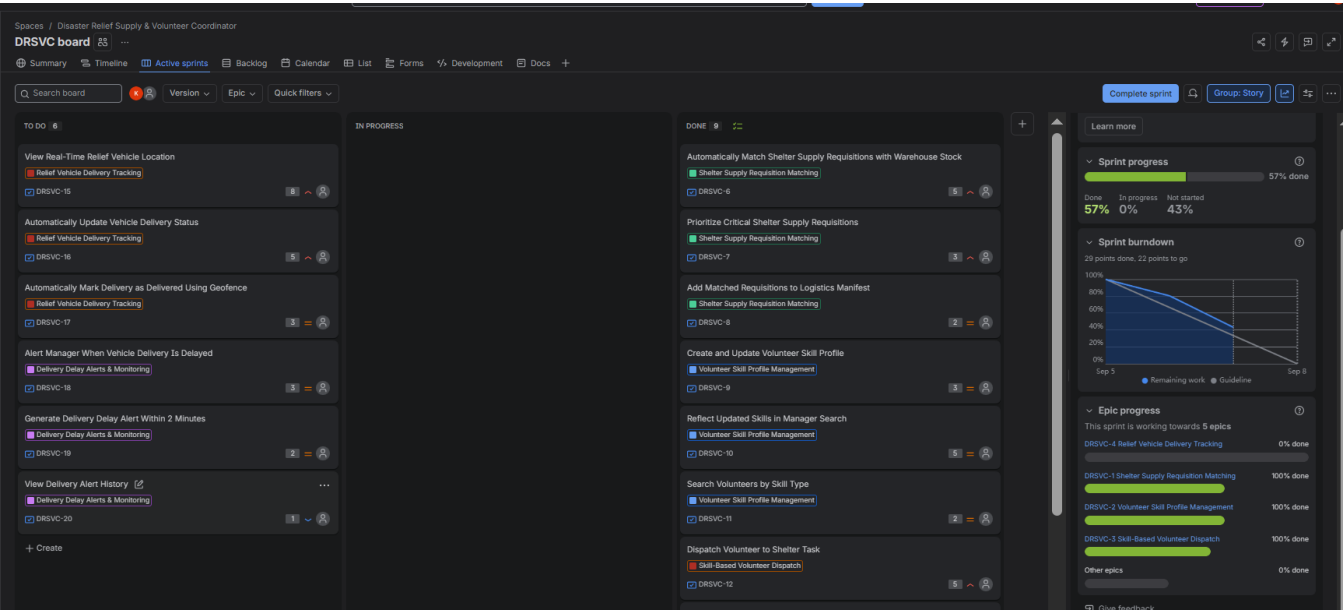


2.2 — Story Point Assignments



2.3 — Sprint Board (Active Sprint view)





2.4 — Burndown Chart



Spaces / Disaster Relief Supply & Volunteer Coordinator / DRSVC board / Reports

Burndown Chart

How to read this chart

	Total				51
05/Sep/26 5:13 PM	DRSVC-6	Burndown	Issue completed	5	46
	DRSVC-7	Burndown	Issue completed	3	43
	DRSVC-8	Burndown	Issue completed	2	41
06/Sep/26 7:13 PM	DRSVC-9	Burndown	Issue completed	3	38
	DRSVC-10	Burndown	Issue completed	5	33
	DRSVC-11	Burndown	Issue completed	2	31
06/Sep/26 7:14 PM	DRSVC-12	Burndown	Issue completed	5	26
	DRSVC-13	Burndown	Issue completed	3	23
	DRSVC-14	Burndown	Issue completed	1	22
08/Sep/26 10:31 AM	DRSVC-15	Burndown	Issue completed	8	14
	DRSVC-16	Burndown	Issue completed	5	9
	DRSVC-17	Burndown	Issue completed	3	6
08/Sep/26 10:32 AM	DRSVC-18	Burndown	Issue completed	3	3
	DRSVC-19	Burndown	Issue completed	2	1
	DRSVC-20	Burndown	Issue completed	1	0
08/Sep/26 6:30 PM	DRSVC-10	Sprint ended by Kari.Bhavitha			0

3. Reflection

1. Did your estimations reflect the actual effort?

Yes, for the most part, my story point estimations were reasonably close to the actual effort required. Smaller stories, such as notifications and viewing delivery alerts, were given fewer points because they were simpler to implement. More complex features, such as real-time vehicle tracking and auto-matching requisitions, were given higher points because they involved more complexity and dependencies. The Fibonacci-based estimation helped me represent the increasing complexity and uncertainty of the larger tasks. Overall, the estimates were fairly accurate, although some real-time or location-based features required slightly more effort than expected.

2. Was your backlog well-prioritized?

Yes, I feel the backlog was well-prioritized based on the importance and urgency of the requirements. Critical disaster-relief functions such as supply matching, volunteer dispatch, and vehicle tracking were given higher priority because they directly affect relief operations. Supporting features such as volunteer skill management and delivery alerts were given medium priority, while less critical features were given lower priority. This helped ensure that the most important functionality was considered first.

3. How did your simulated sprint align with your plan?

The simulated sprint followed the planned Agile workflow. I organized the 15 user stories under 5 epics, assigned story points and priorities, and moved the stories through To Do → In Progress → Done during the sprint. The stories were completed progressively instead of all at once, which made it easier to monitor the progress of the sprint. Overall, the sprint followed the planned workflow reasonably well, although some of the more complex stories required more effort than the simpler ones.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart showed how the remaining workload decreased as the user stories were completed during the sprint. It helped me compare the actual progress with the expected sprint progress and understand how the team's workload was changing over time. The chart also showed that progress was slower in the beginning and increased as more stories were completed. From this, I understood that it is useful to track progress throughout the sprint so that delays can be identified early and future sprint workloads can be planned more realistically.